

Wi-Fi 8 Unveiled: Key Features, Multi-AP Coordination, and the Role of C-TDMA

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ABSTRACT

Wi-Fi 8, the anticipated next generation of Wi-Fi based on the IEEE 802.11bn standard, is designed to meet the growing demands of high-reliability and low-latency applications, such as real-time video, virtual reality, and industrial Internet of things (IoT) communications. This article explores the key technical advances of IEEE 802.11bn, highlighting improvements in the physical and medium access control layers. We emphasize multi-access point coordination as a critical feature for enhancing reliability and reducing latency, specifically coordinated-time division multiple access (Co-TDMA). Performance evaluations of Co-TDMA in various scenarios demonstrate Co-TDMA's effectiveness in optimizing channel access and supporting time-sensitive applications.

INTRODUCTION

With over 21.1 billion connected devices, Wi-Fi remains the dominant wireless technology [1], evolving through successive IEEE 802.11 amendments to meet growing performance demands. Despite competition from fourth generation (4G) small cells and fifth generation (5G) private networks, Wi-Fi continues to lead indoors due to its scalability, cost-effectiveness, and widespread adoption [2].

Each Wi-Fi generation has introduced transformative advancements, evolving from increasing data rates to optimizing system efficiency and reliability [3]. This progression is illustrated in Fig. 1, and summarized below:

- Wi-Fi 4 (802.11n): Introduced 64-quadrature amplitude modulation (QAM) modulation, 4-by-4 downlink (DL) single-user MIMO (SU-MIMO), and packet aggregation, achieving data rates up to 600 Mb/s.
- Wi-Fi 5 (802.11ac): Developed for the 5 GHz band, introducing 160 MHz channels, 80+80 MHz channel bonding, 256-QAM modulation, as well as 8-by-8 DL/uplink (UL) SU-MIMO, and 4-by-4 DL multi-user MIMO (MU-MIMO), reaching peak speeds of up to 6.9 Gb/s.
- Wi-Fi 6 (802.11ax): Boosted performance in dense environments with orthogonal frequency division multiple access (OFDMA), 1024-QAM, and 8-by-8 DL/UL MU-MIMO, delivering maximum throughput of up to 9.6 Gb/s.
- Wi-Fi 7 (802.11be): Enhanced throughput and reliability using 320 MHz channels,

4096-QAM, and multi link operation (MLO), achieving data rates up to 46 Gb/s.

Wi-Fi is now poised to further enhance reliability and reduce latency to address demanding use cases such as real-time video, virtual reality, and industrial Internet of things (IoT) applications [4]. Wi-Fi 8 introduces new mechanisms aimed at improving the responsiveness and predictability of wireless communication, bringing the technology closer to supporting time-sensitive and mission-critical services.

Wi-Fi 8 is being developed under the Institute of Electrical and Electronics Engineers (IEEE) 802.11bn amendment [5], which defines its technical scope and direction. The “bn” designation follows the 802.11 naming convention used in previous amendments, such as 802.11ax for Wi-Fi 6 and 802.11be for Wi-Fi 7.

This article reviews the main objectives and technical features of 802.11bn, examining its benefits and challenges. We emphasize AP coordination as fundamental to Wi-Fi 8's improvements in reliability and latency, and present new system-level simulations demonstrating the potential of Co-TDMA. This work serves as a comprehensive reference for researchers and industry professionals.

IEEE 802.11BN OBJECTIVES AND TIMELINE

The Wi-Fi community continues to advance wireless connectivity through the development of the 802.11bn amendment, which defines the technical foundation of the upcoming Wi-Fi 8 standard. Also referred to as ultra high reliability (UHR) Wi-Fi, 802.11bn is being developed by the IEEE 802.11 working group (WG), which oversees the standardization of Wi-Fi's medium access control (MAC) and physical (PHY) layers. To define the amendment's objectives, key performance indicators (KPIs), supported frequency bands, and candidate technologies, the 802.11bn study group (SG) was formed in Q3 2022. This led to the approval of the project authorization request (PAR) in Q1 2023, followed by the creation of the 802.11bn task group (TG) in Q2 2023 to oversee the technical development.

The amendment aims for a 25% increase in data rates, 25% lower tail latency, and 25% fewer MAC protocol data unit (MPDU) losses, even under mobility and dense overlapping basic service set (OBSS) scenarios [6]. Additional goals include power-saving mechanisms and peer-to-

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peer operations, supporting carrier frequencies from 1 to 7.25 GHz.

802.11bn is expected to be finalized by 2028. Draft D0.1 [5] was completed in Q1 2025, laying the groundwork for the upcoming D1.0 (Q3 2025) and the final D4.0 (Q2 2027) drafts. Final approval by the IEEE 802.11 WG is scheduled for Q1 2028, paving the way for the first Wi-Fi 8-certified products later that year. The Wi-Fi Alliance will certify high-impact features to meet evolving market needs.

PHYSICAL LAYER (PHY)

The 802.11bn TG is actively shaping Wi-Fi 8's core enhancements. This section examines key PHY advancements outlined in the first 802.11bn draft [5].

DISTRIBUTED RESOURCE UNIT

Previous Wi-Fi generations (802.11ax, 802.11be) use resource units (RUs) that consist of contiguous groups of subcarriers. Distributed resource units (DRUs) by contrast define RUs, in which the subcarriers are spread over a larger bandwidth, leading to a lower density of used subcarriers. The main purpose of this is to allow for higher transmit power in power spectral density (PSD)-limited environments. Such DRUs also tend to be more robust in environments with high channel fading. This is especially relevant in the U.S., where FCC rules limit PSD to 5 dBm/MHz for low-power indoor devices in the 6 GHz band (e.g., U-NII-5 to U-NII-8).

DRUs are currently defined for use in UL OFDMA only. The trigger frame used to solicit multi-user UL transmissions has been updated to include signaling for DRUs. The DRU framework is built on a modular architecture, using 26-tone units as the fundamental building blocks. The bandwidth over which a DRU is spread, referred to as distribution bandwidths (DBWs), ranges from 20 MHz to 80 MHz. Larger size DRUs, such as 52-tone, 106-tone, 242-tone, and 484-tone DRUs, are constructed by combining smaller units. DRU tone plans use shifted versions of base configurations to provide uniform performance in fading channels, while tailored long training field (LTF) sequences with low peak-to-average power ratio (PAPR) improve transmission efficiency and reduce distortions. DRUs rely on global cyclic shift diversity (CSD) for spatial stream allocation, helping to mitigate unintentional beamforming.

Challenges: The envisioned transmit power gain depends on the spreading bandwidth of the DRU. To get a significant power benefit, a larger number of DRUs, and hence a larger number of UL stations (STAs), is required. However, it may not always be possible in practice to have the required number of simultaneous STAs available.

ENHANCED LONG RANGE EXTENSION

The enhanced long range (ELR) feature addresses the link budget imbalance between UL and DL transmissions. APs often transmit at higher power levels than STAs, creating UL signal limitations. This imbalance is further exacerbated by regulatory constraints on STA transmit power. ELR improves UL range and reliability, and is thus particularly useful in scenarios where Wi-Fi coverage extends beyond typical indoor environments, such as outdoor campuses, industrial sites, and low-data-rate applications, like IoT and telemetry.

The ELR physical layer protocol data unit (PPDU)

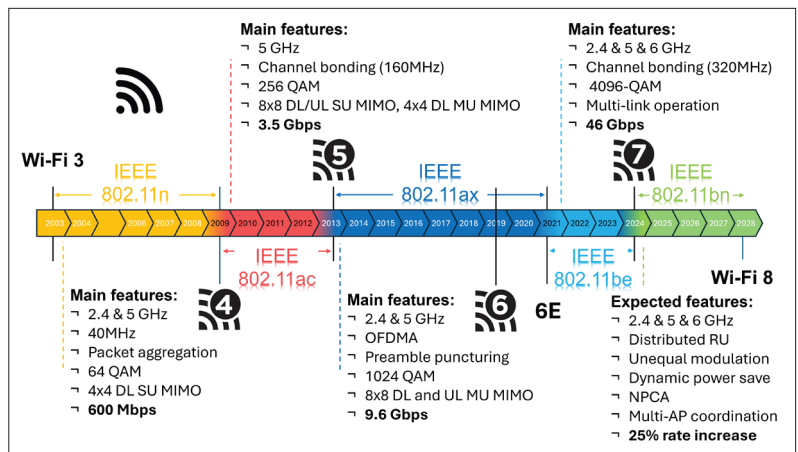


FIGURE 1. Wi-Fi evolution.

is configured for single-user transmission over a 20MHz channel, supporting UHR modulation and coding schemes (MCSs) 0 and 1. It uses a 208-tone RU, applying either $2\times$ or $4\times$ frequency domain repetition over 48 or 96 data subcarriers, respectively. This redundancy improves robustness and reception in challenging UL scenarios. ELR operates in the 2.4 GHz band for both UL and DL, and in the 5 GHz and 6 GHz bands for UL only. This configuration mitigates UL signal limitations while supporting data rates higher than those of legacy direct sequence spread spectrum (DSSS) systems.

To allow for better detection at the receiver, the transmitter of an ELR PPDU is expected to synchronize its transmit clock to that of the AP, typically by compensating for frequency offset based on packets received from the AP.

Challenges: Since ELR targets remote transmitters, the low signal power levels may increase susceptibility to interference, further complicating detection and signal recovery. Moreover, low UHR MCSs rates result in higher airtime consumption, even for small payloads, which may lead to network congestion in dense deployments. In addition, the use of legacy preambles in ELR PPDU introduces substantial overhead, further reducing spectral efficiency, particularly in scenarios with short payloads. Initial preamble detection may benefit from clock synchronization between the ELR transmitter and the AP, but achieving the required accuracy may not always be feasible. The impact of imperfect synchronization on detection performance remains uncertain and requires further evaluation. Introducing ELR-MARK symbols demands a new detection scheme that was not previously part of preamble processing.

LDPC ENHANCEMENTS

The low-density parity check (LDPC) enhancement in Wi-Fi 8 introduces a longer block-length LDPC code of 3888 bits (2×1944), complementing the existing 648, 1296, and 1944-bit code lengths. This enhancement improves error correction performance, addressing UHR requirements in high-reliability scenarios such as extended reality (XR) applications and dense network deployments. Longer codes provide increased coding gain, enhancing performance across various MCS levels.

The 2×1944 LDPC code structure follows a quasi-cyclic (QC-LDPC) design, with a parity-check matrix derived by lifting the base 1944-

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bit codeword. It supports code rates of 1/2, 2/3, 3/4, and 5/6, seamlessly integrating with the updated PPDU encoding parameter table, while maintaining compatibility with legacy codeword lengths for smaller payloads. This modular design enables parallel encoding and decoding, allowing higher throughput without significantly increasing computational complexity. Performance evaluations indicate gains of 0.5 to 1.0 dB, with larger improvements observed for higher MCS levels and multiple spatial streams.

Challenges: With the integration of 3888-bit LDPC codes, managing decoding latency for short payloads is critical, as puncturing can degrade performance. The definition of the 3888-bit codeword relies on the ability to parallelize the existing decoders. Depending on implementation, multiple decoders may already be present in the 5 GHz and 6 GHz bands, limiting the additional overhead. In the 2.4 GHz band, where the widest available channel is 40 MHz, multiple decoders may not be present. To avoid added complexity, it may be advisable to keep the support of 3888-bit codewords limited to the 5 GHz and 6 GHz bands.

UNEQUAL MODULATION AND FINER MCS

In previous Wi-Fi generations utilizing multi-stream multiple-input multiple-output (MIMO) transmissions, all streams operate with the same modulation order and a common code rate. However, in cases where beamforming is used, wireless channels often exhibit significant signal to noise ratio (SNR) variations across spatial streams. Unequal modulation (UEQM) allows to assign different modulation orders to individual streams within a transmission, while joint LDPC encoding ensures consistent error correction across all streams.

By adapting modulation orders to real-time channel conditions, UEQM has the potential to enhance spectral efficiency and reliability, making it particularly beneficial in high-interference or spatially diverse environments. To further accommodate variable link conditions, 802.11bn introduces several new MCSs, providing finer granularity for robust link adaptation. Newly proposed MCSs, i.e., QPSK-2/3 and 16-QAM-5/6, offer greater flexibility in adjusting to diverse scenarios, ensuring optimized throughput and stability.

Challenges: While the additional MCS and UEQM may allow fine-tuning of the modulation choice to optimize performance, the newly available choices for MCS and number of spatial stream combinations lead to additional complexity in rate adaptation, as a larger search space must now be explored. Possibly, rate adaptation could benefit from enhanced feedback mechanism to guide the transmitter in its choice of transmission parameters (MCS among them).

INTERFERENCE MITIGATION

Rate adaptation selects the best transmission parameters for the given channel and noise conditions. However, rate adaptation is too slow to deal with rapidly time-varying interference. 802.11bn is looking at incorporating receiver-based mechanisms to counter such interference. Interference mitigation (IM) enhances interference estimation and suppression by introducing additional pilots within the data portion of the PPDU. These tones can be used to obtain information about the

nature and characteristics of the observed interference. Specifically, when the number of receiver antennas exceeds the number of spatial streams in a received packet, IM pilots can be used for noise covariance estimation, and to configure receiver-side beamforming. The Rx beamforming can project out part or all of the observed interference.

The number of pilot tones needs to be balanced between the need to have sufficient coverage of the spectrum versus the performance loss that results from not using these tones for carrying data.

Challenges: The implementation of IM poses challenges in signaling and configuration. Fixed configurations with predetermined pilot positions are straightforward but lack adaptability. Flexible configurations, which provide predefined alternatives, offer a balance between adaptability and complexity. Fully configurable modes maximize flexibility, but introduce high signaling overhead and implementation challenges. For practical deployment, fixed or flexible configurations are generally preferred. Additionally, IM may be less effective in scenarios such as DL MU-MIMO, where limited receiver resources can restrict interference suppression performance.

MEDIUM ACCESS CONTROL

This section reviews key MAC advancements, their benefits, and challenges from the latest 802.11bn draft [5].

SEAMLESS ROAMING

Seamless roaming in Wi-Fi 8 allows non-AP STA to switch smoothly between APs in the same mobility domain without interrupting ongoing data sessions. This capability is enabled by the multi-link device (MLD) architecture, where a non-AP MLD can maintain simultaneous connections with multiple AP MLDs to support “make-before-break” handovers. This feature is particularly advantageous for time-sensitive applications like video calls, online gaming, and streaming, where a stable and continuous connection is essential, without losing packets during the process.

By preserving critical operational contexts — such as security keys and quality of service (QoS) settings — seamless roaming ensures that performance and security are maintained throughout the transition. However, preserving the security context may introduce risks in some deployments; thus, key renegotiation mechanisms are also supported as secure alternatives. Using a “make-before-break” mechanism, links with the target AP are established before disconnecting from the current AP. Both links remain active during the transition, allowing the STA to receive buffered data from the current AP while forwarded or new packets arrive from the target AP. This approach minimizes the risk of dropped packets, and added redundancy can further enhance reliability during the switch.

Challenges: Security is a key consideration, requiring mechanisms that transfers the security context without compromising privacy. Transferring critical operational contexts between APs also requires careful coordination. Deciding whether to forward data to the target AP or retain it at the source is critical, as is determining the optimal moment to disconnect the link with the current AP.

DYNAMIC POWER SAVE

Dynamic power save optimizes power efficiency for both APs and STAs by enabling transitions between low-capability and full-capability operational modes based on activity, thereby reducing the energy consumption. This feature is particularly beneficial in scenarios that involve sporadic data exchange.

In low-capability mode, devices conserve energy by operating with minimal bandwidth (e.g., 20 MHz), a single spatial stream, and basic MCSs (e.g., non-HT duplicate formats at 6 Mbps or 12 Mbps). Transitions out of the low-capability mode are triggered by an initial control frame (ICF), which includes a padding field to provide the receiving device with sufficient time to reconfigure its hardware for higher-capability operation, without necessarily conveying detailed configuration parameters. Devices should remain in low-capability mode as long as possible, minimizing unnecessary power usage. Importantly, operational modes can also be scheduled. Studies indicate that APs and STAs can reduce power consumption by approximately 50–80% in low-capability mode compared to continuous full-capability operation, depending on device type and traffic patterns.

Challenges: The use of padding in ICFs ensures reliable transitions but introduces additional latency, which can impact delay-sensitive applications. Furthermore, reliance on intermediate frames and additional control overhead may reduce overall efficiency. Integrating these advanced mechanisms with legacy devices adds complexity, requiring seamless adaptation to ensure older systems continue to operate without performance degradation.

NON-PRIMARY CHANNEL ACCESS

Until now, Wi-Fi transmitters are not allowed to access the channel when their primary 20 MHz is occupied. Nonprimary channel access (NPCA) enhances channel utilization by allowing a basic service set (BSS) to initiate channel access via a secondary channel when its primary channel is occupied by OBSS traffic. This capability is particularly valuable where high-bandwidth Wi-Fi 8 nodes coexist with legacy devices with narrower bandwidths, ensuring a more efficient spectrum utilization. Applications requiring low latency also benefit from NPCA by avoiding primary channel congestion.

NPCA operates by enabling both APs and STAs to switch to a predefined NPCA primary channel, with transitions triggered by conditions such as the detection of long OBSS transmissions or request-to-send (RTS)/clear-to-send (CTS) frame exchanges on the primary channel. Critical parameters, including switching delays, may be communicated between devices to synchronize channel access effectively. NPCA is only activated when the opportunity to operate on the non-primary channel is deemed worthwhile, e.g., exceeding a predefined time threshold. Once the frame exchanges on the NPCA primary channel are concluded, both the AP and the STA return to the primary channel based on the NAV of the OBSS transmission that initially triggered the switch to the non-primary channel, ensuring fairness with OBSSs.

Challenges: A key challenge for NPCA is coordinating APs and STAs in hidden node scenarios. If either side misses parts of the RTS/CTS sequence, this can prevent switching to the NPCA

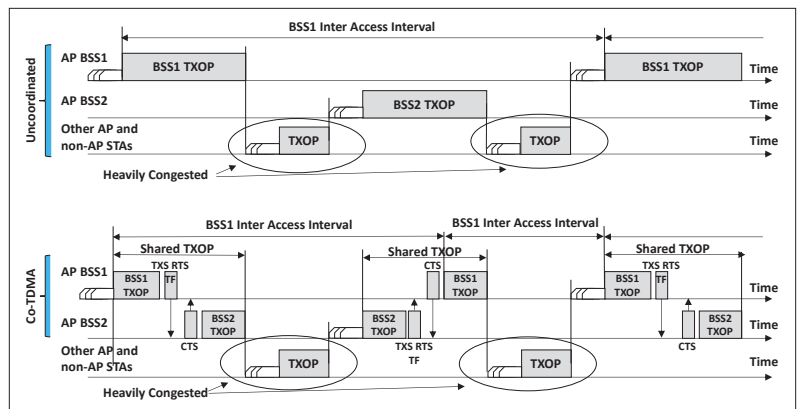


FIGURE 2. Co-TDMA frame exchange.

primary channel or cause mismatched transitions where only one device switches, hindering the efficiency of NPCA. Additionally, not all devices in a BSS may have NPCA capabilities, resulting in fairness issues within the BSS.

MULTI-AP COORDINATION

802.11bn aims to significantly improve reliability and low-latency by enabling coordination among neighboring APs, allowing for more efficient use of spectrum resources. Below, we outline the four coordination approaches being considered in 802.11bn. Although some of them were discussed during 802.11be standardization, they were not formally specified.

Co-TDMA: Co-TDMA enables neighboring APs to share transmission opportunities (TXOPs) in a time-division multiplexing manner, dynamically adapting to each AP's data demands and buffer status (Fig. 2). By facilitating more frequent medium access for time-sensitive applications in congested environments, it may reduce latency on both BSSs.

The mechanism designates a “sharing AP” that allocates portions of its TXOP to one or more “shared APs” within its range [7, 8]. A shared TXOP in Co-TDMA is generally assigned to a specific access category (AC), and the sharing AP must adhere to the requirements of that AC. The coordination process begins with a polling and announcement phase, during which the sharing AP may collect information on the traffic requirements of potential shared APs. Based on this input, the sharing AP dynamically assigns TXOPs, allowing shared APs to utilize their allocated time while returning any unused portions to the sharing AP to avoid resource wastage.

Challenges: Achieving equitable resource allocation among APs and resolving disputes over unused TXOPs demand sophisticated coordination protocols. Deploying Co-TDMA requires extensive enhancements to MAC operations, including precise tuning of the NAV to ensure fairness and prevent collisions. In scenarios where OBSSs handle different types of traffic with varying priorities, the opportunities for sharing of TXOPs can be limited, leading to potential performance issues.

Coordinated Restricted Target Wake Up Time (Co-RTWT): It builds upon the restricted target wake time (R-TWT) feature introduced in Wi-Fi 7. While target wake time (TWT) allowed devices within a BSS to negotiate specific wake times to conserve energy and reduce contention, R-TWT extended this

The sharing AP dynamically assigns TXOPs, allowing shared APs to utilize their allocated time while returning any unused portions to the sharing AP to avoid resource wastage.

by reserving contention-free service periods (SPs) for time-sensitive traffic. Co-RTWT further expands the feature by allowing multiple APs to align or otherwise coordinate their R-TWT schedules [9].

Specifically, during these SPs, non-intended devices terminate ongoing transmissions — TXOPs — to ensure exclusive access for intended devices. APs advertise the scheduled SPs in beacon frames, including parameters such as start time, duration, and membership rules, allowing STAs to join and adjust their transmissions to align with the reserved SPs. A key enhancement in Co-RTWT is support for multi-BSS scenarios, enabling OBSSs to coordinate schedules. Additionally, APs can negotiate schedule alignment dynamically to avoid collisions, leveraging adaptive techniques such as transmission window adjustments and quiet periods tailored to the network density.

Challenges: Overprotection, where frequent — or unnecessarily long — SPs in a BSS lead to APs or STAs in an OBSS to prematurely end their transmissions, can greatly reduce efficiency. Time synchronizing APs is crucial to prevent mismatched schedules that may cause overlapping SPs and collisions. Additionally, hidden nodes that fail to read beacon frames with R-TWT advertisements can cause unintended interference during R-TWT SPs.

Coordinated Spatial Reuse (Co-SR): In Wi-Fi 8, Co-SR builds on the spatial reuse capabilities introduced in Wi-Fi 6, advancing the ability to support simultaneous transmissions within OBSSs through enhanced interference coordination. This advancement is intended not only to reduce latency by minimizing contention but also to increase overall system throughput via improved spectrum reuse [10].

Accurate knowledge of interference is central to Co-SR, as it enables overlapping transmissions to occur without performance degradation [11]. Wi-Fi 8 leverages advanced interference modeling and adjustments based on channel state information (CSI) to dynamically optimize parallel transmission scheduling, resulting in improvements in aggregate throughput. Co-SR is currently based on a TXOP-based reuse approach, which facilitates dynamic and opportunistic concurrent transmissions by providing a framework for APs to negotiate parameters such as interference thresholds, transmit power, and modulation schemes.

Challenges: TXOP-based reuse in Co-SR requires accurate and timely coordination among APs, which introduces overhead due to the need for exchanging interference thresholds, channel state information, and other control parameters. This challenge is amplified in dense and dynamic environments, where interference conditions evolve rapidly, and maintaining precise interference management becomes increasingly complex.

Coordinated Beamforming (Co-BF): Co-BF seeks to reduce latency and enhance throughput by utilizing the antenna domain to direct signal energy toward intended STAs while attempting to eliminate interference to OBSS devices.

Technically, Co-BF leverages MIMO degrees of freedom and digital precoding informed by CSI to form beams that direct signal energy to target STAs and nulls that suppress interference at unintended receivers, with the capability to create as many beams and nulls as the number of antennas available at the AP [12]. It may operate in two modes: full nulling, which eliminates all OBSS

interference but may limit beamforming gain, and partial nulling, which focuses on dominant interference to balance in-BSS performance. New feedback mechanisms building on joint and sequential null packet (NDP)-based sounding can enable a more efficient CSI exchange [13], allowing APs to coordinate transmissions and reducing overhead. Solutions such as limiting Co-BF to two APs are being considered to ensure scalability.

Challenges: For Co-BF, deciding on the optimal number of beams and nulls to schedule, as well as their placement across nodes, is crucial. Effective implementation requires advanced scheduling strategies, such as grouping STAs based on spatial compatibility for MU-MIMO and null steering, to maximize spatial reuse and minimize interference. Additionally, ensuring timely and accurate CSI exchange, including data from OBSS nodes, maintaining precise synchronization of guard interval (GI) during multi-AP transmissions, and reducing feedback overhead are essential to preserve efficiency. The channel information that is needed to enable Co-BF may require frequent soundings of the channel, which increases the overhead of the Co-BF protocol. This overhead needs to be weighed against the potential gains that can be achieved from reducing the interference through beamsteering.

New features such as seamless roaming, NPCA, and multi-AP coordination raise security concerns, requiring robust authentication, key management, and privacy safeguards in future amendments.

C-TDMA PERFORMANCE EVALUATION

Highlighted as one of the main 802.11bn features, this section evaluates Co-TDMA performance using customized, NS3-based system-level simulations in two OBSS scenarios, both reflecting residential-like settings [5, 7, 8].

- **Scenario 1** represents a balanced OBSS setup with four BSSs sharing a single channel, each AP supporting five STAs with uniform traffic types (Fig. 3 and Table 1). Co-TDMA coordination is enabled between BSS 1 and BSS 2, while BSS 3 and 4 remain uncoordinated.
- **Scenario 2** explores unbalanced OBSS conditions, with three BSSs sharing the channel, but differing in the number of associated STAs (7, 3, and 5, respectively) as well as in their traffic loads and ACs (Fig. 3 and Table 1). Co-TDMA coordination is again enabled between BSS 1 and BSS 2, while BSS 3 operates independently.

A single 80 MHz channel in the 5 GHz band is used with MCS 11, representing an upper-bound configuration for high throughput, low latency scenarios in high-SNR indoor environments. This choice allows us to evaluate the benefits of Co-TDMA coordination in densely deployed networks. The propagation model assumes free-space conditions, with all nodes positioned within both energy and packet detection range, ensuring full mutual interference and coordination.

Each STA is assigned one of five service types, differing in throughput, packet size, latency sensitivity, and protocol. Latency-sensitive applications such as video conferencing, gaming, and camera surveillance are modeled using UDP traffic, while high-throughput services like 4K video streaming and cloud file synchronization use TCP. The traffic characteristics for each service type and their mapping to specific STAs are detailed in Table 1.

All nodes employ single-user transmissions.

TXOP SHARING STRATEGIES

Efficient TXOP time-sharing is essential for fairness and latency reduction. We evaluate three TXOP-sharing strategies with varying complexity:

- **Strategy S1 — Post-Buffer-Empty Sharing:** Shares TXOP time only after the buffer of the winning AP is empty, allowing sharing only with same or higher ACs. While simple, it lacks flexibility.
- **Strategy S2 — Soft Time Threshold-Based Sharing:** Relies on a preconfigured time threshold to guide when to release the remaining TXOP. If high-priority traffic finishes early, the sharing AP can voluntarily yield the rest of the TXOP; if a frame is still being transmitted, it may continue briefly beyond the threshold. This method improves efficiency without requiring explicit coordination with other APs.
- **Strategy S3 — Dynamic Buffer-Aware Sharing:** This strategy dynamically determines the duration of TXOP sharing based on real-time buffer occupancy and anticipated traffic demand (via traffic predictions of each AP). During a pre-negotiation phase, APs exchange their buffer status and forecasted transmission needs. Based on this information, the sharing AP computes how much of its TXOP to retain and how much to allocate to neighboring APs. If the combined demand exceeds the TXOP, prioritization rules are applied, typically favoring the sharing AP. While this method is more responsive to varying traffic, it introduces coordination overhead and relies on accurate traffic prediction.

DISCUSSION OF RESULTS

In the following, we analyze the performance of Co-TDMA under the two presented OBSS scenarios. Note that Fig. 4a depicts the 95%-tile round trip time (RTT) latencies for gaming services (Voice (VO) AC), while Fig. 4b illustrates the average throughput rates for video streaming services (Video (VI) AC). The results for Scenario 1 correspond to the four groups of bars at the bottom of each figure, while the results for Scenario 2 are shown by the three groups of bars at the top.

Scenario 1: The key insights derived from this scenario are as follows:

- **Gaming latency:** In coordinated BSS 1 and BSS 2, Co-TDMA reduces gaming latency by around 25% across strategies S1 to S3, with no major differences among them. This improvement stems from efficient TXOP sharing, where the short frame durations enable seamless allocation without timing constraints. Additionally, balanced traffic loads and aligned ACs facilitate both BSS 1 and BSS 2 to access the shared channel frequently and equitably, maximizing TXOP utilization. Meanwhile, uncoordinated BSS 3 and BSS 4 remain largely unaffected, as they maintain approximately equal channel access opportunities, resulting in latency levels comparable to — or slightly larger than — the non-coordinated benchmark. In practice, a misconfigured or malicious AP could exploit longer TXOPs to benefit its group, potentially reducing access for uncoordinated BSSs. This highlights the need for strict TXOP enforcement.
- **Video streaming throughput:** Strategy S1 provides no gains due to its restrictive behavior — TXOPs are only shared after an AP completes its own transmissions, significantly reducing sharing opportunities. Given the larger packet sizes of video streaming traffic, this selfish approach often prevents TXOP sharing altogether, limiting throughput improvements. In contrast, coordinated BSS 1 and BSS 2 achieve 12–17% higher

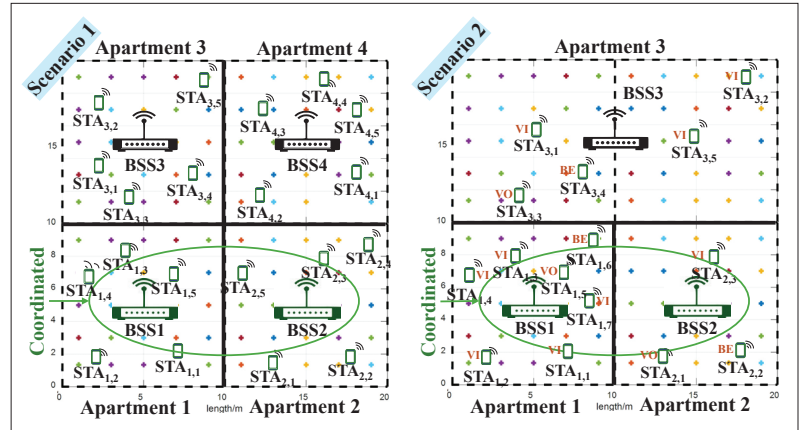


FIGURE 3. Scenario 1 (Scenario 2) with 4 APs (3 APs) and 5 STAs (7,3 and 5 STAs) per AP, each with a different service type as per Table 1.

AP Services	Traffic Characteristics	Scenario	BSS / STA
Video Conference	DL/UL: 3 Mb/s, 250B, [UDP, VI_AC]	Scenario 1	STA _{1,1} , STA _{2,1} , STA _{3,1} , STA _{4,1}
		Scenario 2	STA _{1,1} , STA _{1,2} , STA _{3,1}
Video Streaming (4K)	DL: 32 Mb/s, 1500B, [TCP, VI_AC]	Scenario 1	STA _{1,2} , STA _{2,2} , STA _{3,2} , STA _{4,2}
		Scenario 2	STA _{1,3} , STA _{1,4} , STA _{3,2}
Gaming	DL/UL: 140 kbps, 7 ms periodic, 110B, [UDP, VO_AC]	Scenario 1	STA _{1,3} , STA _{2,3} , STA _{3,3} , STA _{4,3}
		Scenario 2	STA _{1,5} , STA _{2,1} , STA _{3,3}
Cloud File Sync	DL/UL: 10 Mb/s, 1500B, [TCP, BE_AC]	Scenario 1	STA _{1,4} , STA _{2,4} , STA _{3,4} , STA _{4,4}
		Scenario 2	STA _{1,6} , STA _{2,2} , STA _{3,4}
Camera	UL: 2 Mb/s, 1450B, [UDP, VI_AC]	Scenario 1	STA _{1,5} , STA _{2,5} , STA _{3,5} , STA _{4,5}
		Scenario 2	STA _{1,7} , STA _{2,3} , STA _{3,5}

TABLE 1. Traffic generation for each AP/STA in OBSS scenarios.

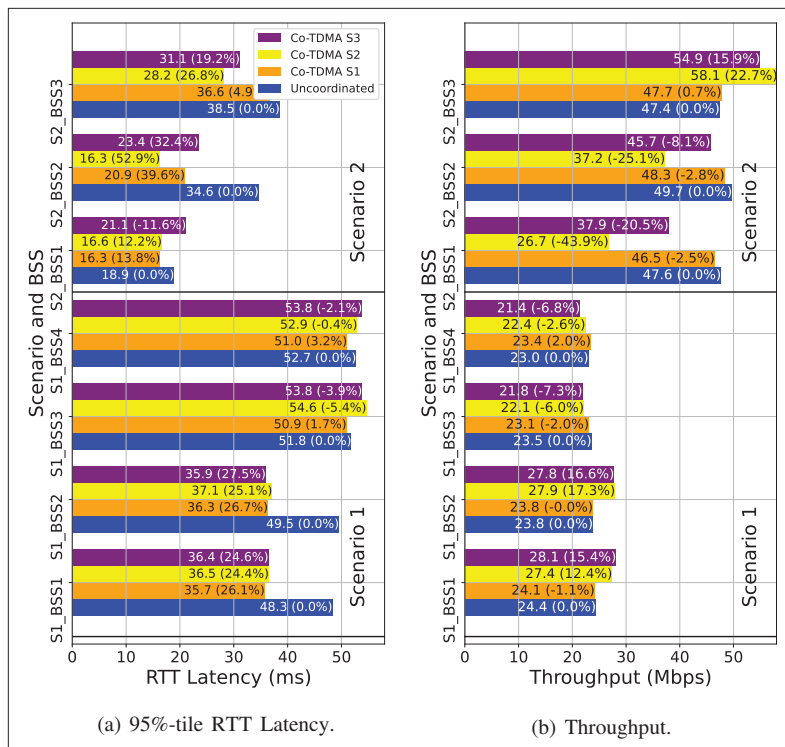


FIGURE 4. Comparison of Gaming 95%-tile RTT Latency and Video Streaming Throughput.

throughput under strategies S2 and S3, primarily due to reduced latency in transmission control protocol (TCP)-based applications. Lower latency and fewer retransmissions allow TCP-based video streaming to sustain higher data rates. Strategy S3 provides no significant advantage over S2 (only 2% improvement on average). Notably, throughput in uncoordinated BSS 3 and BSS 4 remains mostly unaffected.

These results demonstrate that, under balanced traffic conditions, Co-TDMA significantly reduces latency (25%) and increases throughput (12–17%) by facilitating coordinated, fair access sharing across all strategies.

Scenario 2: In this scenario, where OBSSs have differing AC assignments, gaming and video streaming performance are affected accordingly. The key findings are as follows:

- **Gaming latency:** In the uncoordinated case, BSS 1 experiences the lowest gaming latency due to its higher proportion of STAs using the VI AC, which enables more frequent channel access to gaming, which uses the higher VO AC. With coordination, latency generally decreases across all strategies compared to the uncoordinated case. However, the benefit for BSS 1 is limited (up to a 13% reduction) since it was already having a good performance and must now share its frequent channel access. In contrast, BSS 2 significantly benefits from BSS 1's shared TXOPs, achieving a notable latency reduction of up to 53%. Among the coordination strategies, strategy S2 outperforms S1 and S3 because it forces TXOP sharing. Strategy S3 underperforms in terms of latency due to its reliance on estimated TXOP availability, which is low in this scenario. Since BSS 2 has little VI/VO traffic,

it generates few opportunities to share with BSS 1. Moreover, due to its low traffic, BSS 2 also estimates minimal sharing needs, causing BSS 1 to rarely share. Additionally, the estimator may struggle in this scenario because it relies on average estimations, while the traffic generator introduces jitter, further reducing efficiency. Notably, BSS 3 also benefits from the better performance of the other BSSs, achieving up to a 26% latency reduction due to increased channel access opportunities and reduced collisions.

- **Video streaming throughput:** Video streaming throughput suffers in this scenario due to insufficient coordination between BSS 1 and BSS 2. Under strategy S1, the impact on throughput is minimal, as the strategy is selfish, only sharing resources when surplus TXOP time is available. However, strategy S2 experiences more pronounced degradation, with throughput reductions of up to 44% in BSS 1 and 25% in BSS 2. S2 performs poorly due to its strict resource allocation. If a scheduled TXOP is not fully utilized, the excess time is wasted rather than reassigned, resulting in significant throughput losses. BSS 2 is less affected, as it is able to capitalize on the more frequent sharing opportunities from BSS 1. This inefficiency is partially mitigated in S3, which adapts more dynamically and reduces throughput loss (up to 20% reduction in BSS 1 and 8% in BSS 2). Meanwhile, BSS 3 benefits from the performance degradation of BSS 1 and BSS 2, as reduced competition for channel access allows it to achieve throughput gains of up to 22%.

These results highlight the importance of aligned traffic priorities and balanced loads for effective Co-TDMA operation. Under unbalanced or misaligned conditions, Co-TDMA yields inconsistent gains, diminishing the incentive for cooperation.

CONCLUSION

In this article, we presented the main features of the upcoming Wi-Fi 8, focusing on both the PHY and MAC layers. Leveraging Co-TDMA under the right conditions, Wi-Fi 8 can significantly improve reliability and reduce latency, making it ideal for time-sensitive applications. Performance evaluations show a 25% reduction in latency and up to 17% higher throughput in OBSS scenarios. These results demonstrate Wi-Fi 8's potential to meet the demands of next-generation wireless applications.

While this work focused on Co-TDMA due to its deterministic benefits for time-sensitive traffic, other coordination mechanisms — such as Co-RTWT, Co-SR, and Co-BF — offer different performance trade-offs. A comparative analysis of these alternatives is a valuable direction for future work.

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